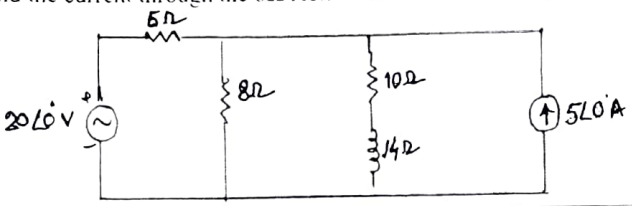
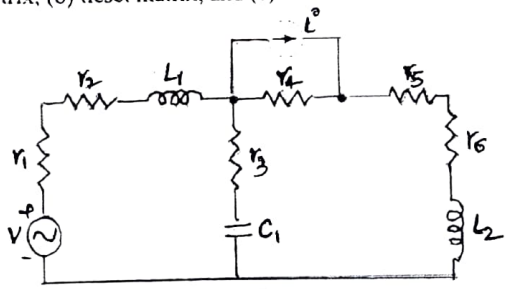
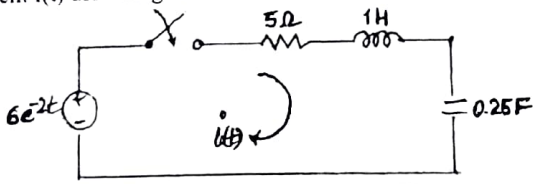
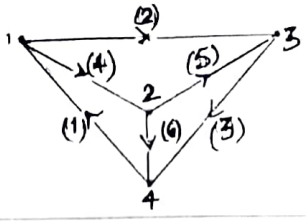
MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

Paper Code: EEC-206 | Subject: Network Analysis and Synthesis
Time: 1½ Hrs. | Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO(s)
1	(a) State Norton's Theorem.	3	1
	(b) Define twigs and links of tree.	2	1
	(c) Find the Laplace transform of e^{t-a}, $t > a$	3	2
	(d) Define transient and steady state Response.	2	2
2	(a) Find the current through the 8Ω resistor in the network of figure. 	5	1
	(b) For the circuit shown in figure, draw the oriented graph and write (a) incidence matrix, (b) tieset matrix, and (c) cutset matrix 	5	1
3	(a) Find the Laplace transform of $\int_0^t e^{-2t} t^3 \, dt$	5	2
	(b) For the network shown in figure, the switch is closed at t = 0. Determine the current i(t) assuming zero initial conditions in the network elements. 	5	2
4	The graph of a network is shown in figure. Write the (a) incidence matrix, (b) tieset matrix, and (c) f-cutset matrix. 	10	2